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Module 5 Introduction

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Module 5-1 Genetic Algorithm 1

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Module 5-2 Genetic Algorithm 2

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Module 5-3 Use Genetic Algorithm 1

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Module 5-4 Use Genetic Algorithm 2

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Module 5 Quiz

Module 5 Assignment

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Solving Real Engineering Problems with Genetic Algorithms



In this section, you applied a genetic algorithm in MATLAB to solve a real-world antenna design problem by translating frequency-domain performance requirements into an optimization objective.

Reflect on the following questions:

- Why is it necessary to carefully design the **objective function** when solving real engineering problems with GA, rather than directly optimizing raw performance metrics?
- How did **bounds**, **population size**, and **maximum generations** influence the behavior and efficiency of the GA in this case study?
- Why is it important to evaluate GA results using both the **objective value** and the **physical performance response** (such as the S11 curve)?
- Based on this case study, what challenges do you anticipate when applying GA to more complex engineering problems involving simulations?

This discussion will help you consolidate how genetic algorithms are used in practice and prepare you to apply them responsibly in engineering and scientific applications.

✓ Completed

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